

Prolog Program to Implement Pattern Matching

Aim

To write a Prolog program to implement **pattern matching** by comparing two lists and checking whether their elements match.

Algorithm

1. Start.
2. Take two lists as input.
3. Compare the first elements of both lists.
4. If they match, compare the remaining elements.
5. Continue until both lists are empty.
6. If all elements match, return `true`.
7. If any element does not match, return `false`.
8. Stop.

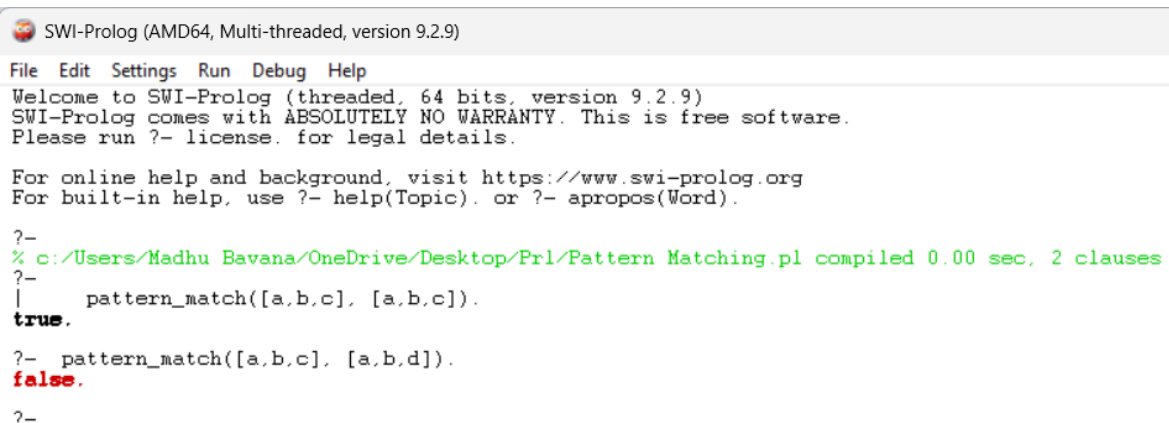
Prolog Code

```
% Pattern Matching

pattern_match([], []).

pattern_match([H1|T1], [H2|T2]) :-
    H1 = H2,
    pattern_match(T1, T2).
```

OUTPUT

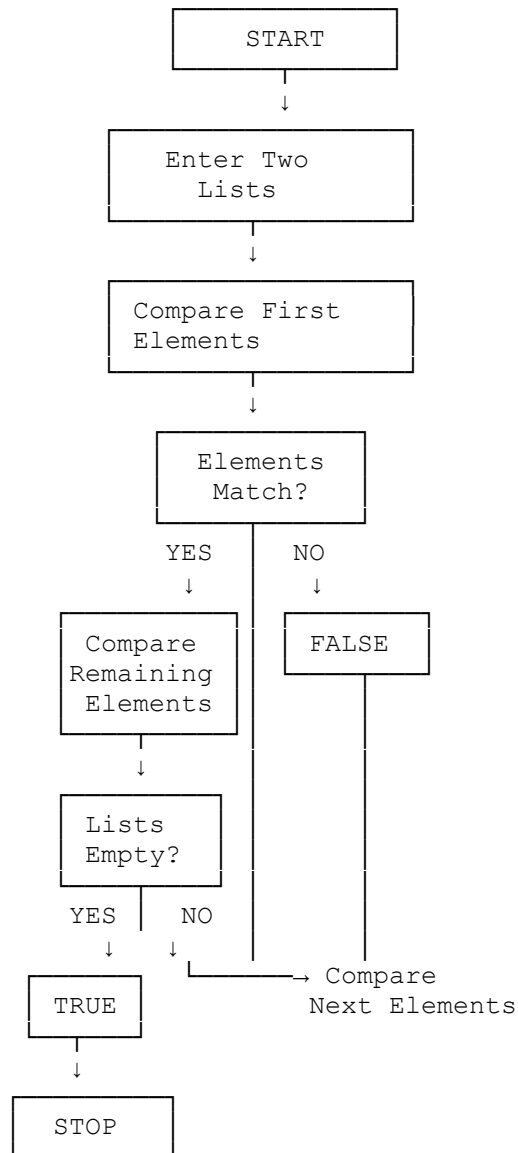


The screenshot shows the SWI-Prolog (AMD64, Multi-threaded, version 9.2.9) environment. The menu bar includes File, Edit, Settings, Run, Debug, and Help. The status bar indicates 'Welcome to SWI-Prolog (threaded, 64 bits, version 9.2.9)' and 'SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software. Please run ?- license. for legal details.' The main window displays the following text:

```
For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

?-
% c:/Users/Madhu Bavana/OneDrive/Desktop/Prl/Pattern Matching.pl compiled 0.00 sec, 2 clauses
?-
|   pattern_match([a,b,c], [a,b,c]).
|   true.
?- pattern_match([a,b,c], [a,b,d]).
|   false.
?-
```

Flowchart



Result

```
?- pattern_match([a,b,c], [a,b,c]).  
true.
```

Thus, the Prolog program successfully performs pattern matching between two lists.